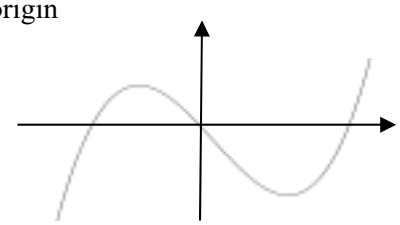
Question	Answer	Marks	Partial Marks
9(b)	14500 or 14470 to 14480	**4**	**M3** for $200 \times 60 \times 24 \times \pi \times 4^2$ [÷1000] or $2 \times 60 \times 24 \times \pi \times 0.04^2$ [×1000] or **M2** for $200 \times \pi \times 4^2$ or for $2 \times \pi \times 0.04^2$ or **M1** for $\pi \times 4^2$ oe or $\pi \times 0.04^2$ seen oe isw or 1000 cm^3 = 1 litre soi or 1 m^3 = 1000 litres soi or for 24 × 60 seen oe
10(a)	$x^3 + 2x^2 - 5x - 6$ final answer	**3**	**B2** for correct expansion of three brackets unsimplified or for simplified expression of correct form with 3 out of 4 terms correct or **B1** for correct expansion of 2 of the 3 given brackets with at least 3 terms out of four correct
10(b)	$\frac{Mc}{M-2f}$ or $\frac{-Mc}{2f-M}$ final answer	**4**	**M1** for clearing $g - c$ from denominator e.g. $M(g - c) = 2fg$ **M1** for correctly isolating terms in g in numerator on one side **M1** for correctly factorising or simplifying, to single term in g in an equation **M1** for correctly dividing by bracket to final answer
10(c)	$\frac{4x}{x+4}$ final answer	**3**	**B1** for $4x(x - 4)$ **B1** for $(x + 4)(x - 4)$
11(a)(i)	$\frac{1}{6}$ oe on all late branches $\frac{5}{6}$ oe on all not late branches	**2**	**B1** for one correct vertical pair $\frac{1}{6}$ oe and $\frac{5}{6}$ oe
11(a)(ii)	$\frac{5}{36}$ oe	**2**	FT *their* tree **M1** for $\textit{their}\,\frac{1}{6} \times \textit{their}\,\frac{5}{6}$
11(b)(i)	$(G \cup T \cup M)'$ oe	**1**	
11(b)(ii)	28	**1**	
11(b)(iii)	$\frac{17}{50}$ oe	**1**	

Question	Answer	Marks	Partial Marks
11(b)(iv)	$\frac{4}{7}$ oe	**3**	**M2** for $\frac{16}{21}\times\frac{15}{20}$ or **M1** for $\frac{n}{21}\times\frac{n-1}{20}$ or for $\frac{16}{21}$ and $\frac{15}{20}$ seen If 0 scored **SC1** for answer $\frac{256}{441}$ oe
12(a)	85[.0], 265[.0] and no others	**2**	**B1** for each If 0 scored **SC1** for two values in the range with a difference of 180 but not multiples of 90
12(b)	correct shape and passes through origin 	**3**	**B1** for any positive cubic shape **B1** for sketch with one max and one min and with 3 roots including zero If 0 scored, **SC1** for $x(x + 2)(x – 2)$ soi
12(c)	$a = –12$ $b = 5$ $k = –11$	**6**	**B5** for 2 correct OR **B2** for $3x^2 + a$ or **B1** for $3x^2$ isw **M1dep** on at least B1 for *their* $\frac{dy}{dx} = 0$ **M1dep** on at least B1M1 for $x = 2$ or $x = – 2$ substituted in *their* $\frac{dy}{dx} = 0$ equation **M1** for $k = 2^3 + 2\times \textit{their}\, a + b$ **and** $10 - k = (-2)^3 + (-2)\times \textit{their}\, a + b$